THE LAST BEACON

A CINEMATIC 2D SCI-FI ADVENTURE

FULL GAME PLAN & STORY ABSTRACT

Pixel Art • Platforming • Scientific Puzzles • Exploration

Project vision

The Last Beacon is a story-driven 2D science-fiction platformer in which players guide molecular biologist and astronaut Ryland Grace from a research campus at SUIIT into deep space. There, he encounters Rocky, an intelligent alien engineer. Together, they investigate a cosmic biological threat and attempt to protect two civilisations.

Design direction: cinematic pixel art with atmospheric lighting, expressive character animation, environmental storytelling and puzzles rooted in science and cooperation.

1. Executive Summary

The Last Beacon is planned as a single-player, 2D side-scrolling adventure with platforming, exploration, survival systems and logic-based puzzles. Its central mechanic is the ability to switch between two characters whose abilities complement one another. Ryland uses scientific analysis and human technology; Rocky uses alien engineering, climbing and physical strength.

The experience begins at SUIIT, establishing a distinctive Earth-based opening before expanding into a damaged interstellar spacecraft, deep space, an alien ecosystem and the Eridian civilisation. The narrative progresses from scientific curiosity to first contact, cooperation, sacrifice and a new sense of home.

Project goals

• **Distinctive setting:** use a recognisable SUIIT-inspired campus as the opening location.

• **Meaningful mechanics:** make science and teamwork the primary tools for progress.

• **Emotional storytelling:** build the relationship between the two protagonists through playable moments.

• **Manageable production:** develop a polished vertical slice before expanding to the complete game.

2. Game Snapshot

Item	Proposed direction
Genre	2D cinematic platformer; puzzle adventure; exploration and light survival
Visual style	Pixel art with cinematic lighting, parallax backgrounds, particles and controlled bloom
Camera	Side-scrolling camera with occasional fixed cinematic compositions
Players	Single-player; switchable playable characters
Characters	Ryland Grace and Rocky
Setting	SUIIT campus, Earth orbit, Hail Mary spacecraft, deep space, alien world and Erid
Target platform	Windows PC initially; controller support can follow
Engine	Godot 4 with GDScript
Target length	Full experience: approximately 3–5 hours; initial demo: 20–30 minutes
Design resolution	1280 × 720 output; 320 × 180 logical pixel-art canvas as a starting point

3. Core Gameplay Pillars

Scientific problem-solving

Players scan objects, collect samples, interpret clues and solve puzzles involving simplified biology, chemistry, physics and engineering.

Two-character cooperation

Ryland and Rocky have different movement, interaction and problem-solving abilities. Certain obstacles require switching between them.

Exploration and survival

Each environment has unique hazards, secrets, resources and environmental clues. Oxygen, power and repair components support selected sequences.

Cinematic emotional storytelling

Dialogue, animation, music and environmental details communicate the growing bond between the characters.

4. Main Characters

Ryland Grace — Human scientist and astronaut. Ryland is the primary scientific problem-solver. His abilities include scanning, biological analysis, operating human technology, using tools and limited jetpack movement.

Rocky — Alien engineer. Rocky provides a different perspective on technology and the environment. His abilities include climbing designated surfaces, moving heavy objects, repairing alien machinery, interpreting sound patterns and accessing narrow passages.

Character switching

The player switches between Ryland and Rocky using a dedicated input (proposed default: TAB). The active character is shown in the HUD. Puzzles should communicate clearly which character or ability is needed, while allowing optional experimentation.

5. Core Mechanics

System	Function in gameplay
Movement	Run, jump, climb where permitted, interact with objects and traverse platforming routes.
Scanner and journal	Ryland scans organisms, machinery and anomalies. Findings are stored as clues and research entries.
Character cooperation	Switch characters to reach different routes, operate paired mechanisms and solve multi-stage puzzles.
Puzzle interaction	Molecular sequencing, signal matching, circuit repair, gravity routing and environmental logic.
Resource management	Selected sequences use oxygen, energy, fuel and repair components. Resources should support tension, not create constant frustration.
Dialogue and translation	Rocky's musical communication becomes progressively understandable through a translation mechanic.
Progression	Research upgrades unlock new scanning, traversal, translation and engineering capabilities.

6. Story Structure and Chapter Plan

Chapter 01 — The Signal from Earth

Location: SUIIT campus and research laboratory

Story: Ryland investigates an unusual meteorite sample and detects a mysterious biological signature. A solar anomaly leads to an emergency recruitment message.

Level sequence: Campus exploration; laboratory interaction; scanner tutorial; molecular puzzle; antenna repair and signal decoding.

Gameplay purpose: The player learns movement, interaction, scanning and basic puzzle logic.

Chapter 02 — The Hail Mary

Location: Earth orbit and spacecraft

Story: Ryland boards the interstellar spacecraft. A crisis damages systems and leaves him disoriented. He must restore essential systems and navigate an asteroid field.

Level sequence: Launch sequence; oxygen emergency; power restoration; navigation puzzle; asteroid-dodging sequence; unknown contact.

Gameplay purpose: Introduces spacecraft survival, resource management and navigation challenges.

Chapter 03 — The Friend from Another World

Location: Deep space and alien spacecraft

Story: Ryland encounters Rocky. They initially cannot communicate, but a translation system reveals that they face a related cosmic threat.

Level sequence: Explore alien ship; decode musical signals; build translator; switch characters; operate paired control systems.

Gameplay purpose: Introduces Rocky, character switching and cooperative puzzles.

Chapter 04 — The Planet of Secrets

Location: Alien ecosystem and research sites

Story: The pair investigate a biological ecosystem to identify a natural predator or countermeasure for the cosmic organism.

Level sequence: Alien traversal; sample collection; cave exploration; environmental hazards; ecosystem and molecular puzzles.

Gameplay purpose: Expands exploration, biological discovery and character-specific traversal.

Chapter 05 — The Cost of Friendship

Location: Deep space and damaged spacecraft

Story: A contamination and mechanical emergency threaten the research and Rocky's ship. Ryland must choose to help his friend rather than prioritise a safe return.

Level sequence: Containment; engine repair; resource collection; rescue sequence; route decision and emotional climax.

Gameplay purpose: Combines previously learned mechanics under pressure and delivers the major emotional turning point.

Chapter 06 — The Last Beacon

Location: Erid and the Eridian civilisation

Story: Rocky returns home with the discovery. Grace helps complete the final experiment and transmit the solution to Earth, then finds a new role among the Eridians.

Level sequence: Explore Erid; repair infrastructure; final biological experiment; activate beacon; transmit discovery; closing reunion.

Gameplay purpose: Resolves the central mission and completes the friendship arc.

7. Opening Storyboard

The opening sequence should establish the scale of the threat and then bring the player into a familiar, grounded environment. The first 90 seconds are designed as a cinematic hand-off into player control.

00:00–00:15 | The Warning

A dark screen fills with stars. Earth appears. A restrained voice-over hints that something is changing in the universe.

00:15–00:30 | SUIIT

The camera moves from Earth to the SUIIT-inspired campus at night. A single research laboratory remains lit.

00:30–00:45 | The Discovery

Ryland examines a meteorite sample. A biological signal appears on the monitor and the laboratory warns of an unidentified organism.

00:45–01:00 | The Mission

An emergency transmission requests Ryland's presence. A star map highlights the distant mission destination.

01:00–01:15 | The Journey Begins

A launch montage follows the spacecraft leaving Earth. Music rises, then drops into silence.

01:15–01:30 | Wake Up

Ryland wakes inside a dark spacecraft as an alarm sounds. The camera settles into the playable side-scrolling view and control is handed to the player.

8. Key Puzzle Concepts

Molecular reconstruction

Scan a sample, identify its components and arrange them in a correct sequence. Used in Chapters 1, 4 and 6.

Rocky's language

Listen to recurring musical patterns, compare their rhythm and frequency, and match them to meanings. Vocabulary builds over time.

Circuit repair

Connect power sources and components to restore systems. Later versions introduce multiple power routes and hazards.

Gravity challenge

Activate or redirect gravity systems to change platform positions and traversal routes.

Dual engineering

Use Ryland at a human console and Rocky at an alien mechanism to synchronise two systems.

Final biological experiment

Combine clues and discoveries from earlier chapters to stabilise the final countermeasure.

9. Side Missions and Collectibles

Optional content rewards exploration without blocking the main story. Suggested collectibles include scientific research logs, alien communication recordings, spacecraft blueprints, biological samples and friendship memories. Side missions may unlock bonus dialogue, journal entries, upgrades or a short post-credit scene.

Suggested side missions

Mission	Chapter	Reward
Hidden SUIIT research logs	1	Background lore
Lost space data	2	Upgrade components
Rocky's musical memory	3	Translation vocabulary
Unknown species	4	Research entries
Emergency repair challenge	5	Additional resources
Eridian knowledge archive	6	Bonus ending scene

PART II — PRODUCTION PLAN

10. Visual and Audio Direction

Visual language: pixel-art characters and tile-based environments supported by cinematic lighting, parallax layers, atmospheric particles and restrained glow effects. Each world should have a distinct palette: warm natural tones for Earth, deep navy and electric blue for space, orange and red for hostile alien terrain, and luminous greens and warm highlights for Erid.

Animation priorities: idle, walk, run, jump, fall, interact, scan, repair and damage states for Ryland; idle, move, climb, lift, repair, signal and reaction states for Rocky. Prioritise readable silhouettes and expressive timing.

Audio: use environmental ambience in each location, mechanical sounds in the spacecraft, original musical motifs for Rocky's communication, tension layers for emergencies and a warmer orchestral palette for the conclusion. Avoid relying on licensed film music.

11. User Interface

• Top-left: active character portrait and health/status.

• Top-right: oxygen, energy or other chapter-specific resources.

• Bottom-left: current objective and optional hint.

• Bottom-right: character-switch indicator and relevant ability prompt.

• Contextual interaction prompt near usable objects; avoid clutter during exploration.

• Research journal and map accessible through pause or dedicated controls.

12. Technology and Project Structure

Engine: Godot 4. **Language:** GDScript. **Art:** Aseprite or Pixelorama. **Tilemap planning:** Tiled or Godot TileMapLayer. **Audio editing:** Audacity. Use version control (for example, Git) and maintain a shared naming convention for scenes, scripts and assets.

Suggested folder structure

```
THE_LAST_BEACON/
■■■ assets/ characters/ environments/ backgrounds/ animations/ sound/ music/ ui/
■■■ scenes/ main_menu/ characters/ chapters/ puzzles/
■■■ scripts/ player/ character_switch/ dialogue/ inventory/ puzzles/ save_system/
■■■ project.godot
```

13. Development Roadmap

Phase	Focus	Key deliverable
Weeks 1–2	Pre-production	Story outline, character concepts, campus map, task plan
Weeks 3–4	Core mechanics	Movement, camera, interactions, basic animation and dialogue
Weeks 5–6	Chapter 1	SUIIT campus, laboratory, first puzzle and opening cinematic
Weeks 7–8	Space systems	Hail Mary interior, oxygen/power systems and navigation challenge
Weeks 9–10	Rocky and switching	Rocky prototype, character switching and cooperative puzzle
Weeks 11–12	Alien environment	Terrain, hazards, samples and biological puzzle

Phase	Focus	Key deliverable
Weeks 13–14	Presentation	Cutscenes, dialogue, audio, menus and HUD
Weeks 15–16	Polish and demo	QA, optimisation, bug fixes and Windows demo build

Planning note

The 16-week schedule is a target for a polished vertical slice, not a guaranteed schedule for the complete 3–5 hour game. Full production should be estimated after the prototype establishes asset complexity, team capacity and level-building speed.

14. Team Roles

Role	Responsibilities
Game director / producer	Scope, narrative consistency, milestones and coordination
Gameplay programmer	Movement, switching, puzzles, interaction and save systems
Pixel artist / animator	Character sprites, environments, effects and animation
Level designer	Campus, spacecraft, alien levels, routes and pacing
UI/UX designer	HUD, menus, journal and accessibility
Audio designer	Ambience, effects, music direction and alien signal language
QA tester	Playtesting, bug reporting, puzzle clarity and performance checks

PART III — DELIVERY AND SUCCESS CRITERIA

15. First Playable Demo (Vertical Slice)

The recommended first release is a 20–30 minute demo that proves the game's visual identity, core movement, scientific puzzle design and character partnership. It should contain three connected environments and end at the first meaningful encounter with Rocky.

Demo sequence

1. SUIIT campus

Explore a small campus area, learn movement and enter the research laboratory.

2. Laboratory discovery

Scan the sample, solve the first molecular puzzle and receive the solar anomaly transmission.

3. Hail Mary

Transition to the spacecraft, restore a critical system and complete a short survival challenge.

4. Unknown contact

Detect and approach the alien ship.

5. First encounter

Meet Rocky, hear his musical signal and complete a simple communication interaction.

Demo acceptance criteria

• Player can complete the demo from start to finish without blocking bugs.

• Movement, collisions, interaction prompts and camera behaviour are reliable.

• The first scientific puzzle is understandable and has clear feedback.

• The story hand-off from SUIIT to space is coherent.

• The first encounter with Rocky creates a memorable ending hook.

• The game runs at a stable frame rate on the target development PC.

16. Risks and Mitigation

Risk	Mitigation
Scope becomes too large	Build a vertical slice first; lock chapter scope before producing all art and levels.
Too many unique assets	Reuse modular tiles, props and lighting setups while giving each world a distinct palette.
Puzzles become confusing	Use environmental clues, progressive hints, clear feedback and playtesting.
Character switching feels awkward	Prototype switching early; make character roles clear and avoid unnecessary switching.
Performance issues	Profile early, limit expensive effects and test on target hardware throughout production.
Intellectual-property concerns	Treat film characters and story as fan-game material; obtain rights or replace them with original characters before public/commercial release.

17. Success Measures

- A new player understands the basic controls within the opening minutes.
- The player uses both characters meaningfully in the demo.
- At least one puzzle combines observation and scientific reasoning.
- The opening establishes SUIIT as a memorable location rather than a brief backdrop.
- The Rocky encounter creates curiosity about the next chapter.
- The demo is stable, completable and suitable for a university project presentation.

18. Closing Concept Abstract

The Last Beacon is a cinematic 2D science-fiction adventure about discovery, cooperation and friendship across worlds. Beginning at SUIIT, the game follows Ryland Grace from a mysterious laboratory discovery into an interstellar mission. His encounter with Rocky transforms the journey from a solitary survival mission into a partnership between two very different beings. Through platforming, exploration, scientific puzzles and environmental storytelling, players uncover a biological mystery and work toward saving two civilisations.

The project's defining feature is its dual-character design: Ryland's scientific reasoning and Rocky's alien engineering must work together to overcome obstacles neither could solve alone. With pixel-art environments, atmospheric lighting, an evolving musical communication system and an emotionally driven chapter structure, The Last Beacon aims to deliver a memorable indie-game experience with a distinctive university-inspired beginning.

THINK. EXPLORE. EXPERIMENT. COOPERATE. SURVIVE.

Document note: This is a concept and planning abstract. Story beats, characters and world references associated with Project Hail Mary are included for an unofficial fan-game concept. Permission or substantial originalisation is required for public or commercial distribution.